

A03 Reflection

In class, we learned about how temperature and top_p can affect the output of a LLM. Temperature affects the deterministic and rigidity of the output. The lower the value, the more robotic and deterministic the answer becomes, but also the more direct it is delivered. The higher the value, the more extra divergence and fluff is added to the response, making it less robotic but at the cost of an increased probability of deviation and lack of coherence. Top_p affects the LLM's choices for each word to use in its answer. A higher value makes it consider all possible word choices when picking each word for its response, a lower value only words with a high probability of being used/relevant. In this case, I believe temperature is possibly the parameter that brought the bot to fail in its calculations. In the code I wrote for this assignment, I set the temperature to 1.0. This was necessary since in order to enable the LLM to write a response in the style of a teacher teaching his student and to be able to write a self-deprecating joke if it failed, it needed to be able to use an expressive vocabulary. However, a temperature setting of 1 adds possibilities of deviation in the bot's response. And in mathematics, where precision is of utmost importance, even the smallest deviation can make an entire equation invalid. And when you must handle very large numbers, there is a greater possibility of committing mistakes. The temperature requirement to enlarge the bot's vocabulary is what I believe to have led it to perform calculation mistakes.